Shujun Zhao

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Education

University of South Florida, Tampa, FL — Aug 2022 - present
Ph.D. in Mathematics, expected August 2027
Advisor: Xiang-dong Hou

Guilin University of Electronic Technology, Guilin, China — Sep 2018 - Jul 2022
B.S. in Information and Computing Science

Research Interests

Finite fields; polynomials and rational functions over finite fields; function fields and Drinfeld modules; applications to coding theory and cryptography.

Publications and Accepted Papers

1. Xiang-dong Hou and **Shujun Zhao**. “On a Conjecture About the Sum-Freedom of the Binary Multiplicative Inverse Function.” Designs, Codes and Cryptography 94, no. 4 (2026), 86.

2. Alyssa Ebeling, Xiang-dong Hou, Ashley Rydell, and **Shujun Zhao**. “On Sum-Free Functions.” Finite Fields and Their Applications 110 (2026), 102744.

3. Xiang-dong Hou and **Shujun Zhao**. “Two Absolutely Irreducible Polynomials over F_2 and Their Applications to a Conjecture by Carlet.” Finite Fields and Their Applications 109 (2026), 102713.

4. Luca Bastioni, Giacomo Micheli, and **Shujun Zhao**. “On the Characteristic Polynomial of Linearized Polynomials.” Accepted for publication in Mathematics of Computation.

5. Mohamed O. Darwish, Anuvrat Jaindungarwal, Giacomo Micheli, Divyesh Vaghasiya, and **Shujun Zhao**. “Good Polynomials for Locally Recoverable Codes: Classification Results, Asymptotics, and Exact Formulae.” Journal of Algebra and Its Applications (2025), 2541017.

Preprints

1. Xiang-dong Hou and **Shujun Zhao**. “Further Results on Sum-Freedom of Binary and q-ary Functions.” arXiv:2609.31489.

2. Xiang-dong Hou, Siyu Peng, Yongyu Qiang, and **Shujun Zhao**. “Classification of Rational Functions of Degree Three over Finite Fields.” arXiv:2604.18954.

Research Experience

Research Assistant, REU Site: Cryptography and Coding Theory
University of South Florida | Supervisor: Xiang-dong Hou

May - August 2025: Investigated the classification of degree-three rational functions over finite fields of odd characteristic.

May - August 2024: Investigated Carlet’s conjecture on the sum-freedom of the binary multiplicative inverse function.

Invited Talks

On sum-freedom of binary power functions. AMS Special Session on Developments in the Theory of Boolean Functions, Joint Mathematics Meetings, Chicago. January 2027 (scheduled).

On a conjecture about the sum-freedom of the binary multiplicative inverse function. AMS Spring Eastern Sectional Meeting, Special Session on Finite Fields and Their Applications, Boston. March 2026.

Conference Presentations

Classification of rational functions of degree three over finite fields. 11th International Workshop on Boolean Functions and their Applications (BFA), Sogndal. September 2026. Presented online.

On a conjecture about the sum-freedom of the binary multiplicative inverse function. 10th International Workshop on Boolean Functions and their Applications (BFA), Cyprus. September 2025. Presented online.

Summer Schools and Professional Development

Graphical Models in Algebraic Combinatorics, Simons Laufer Mathematical Sciences Institute (SLMath). June - July 2025.
Nominated by the Department of Mathematics and Statistics, University of South Florida.

Teaching Experience

University of South Florida, Tampa, FL

Primary Instructor | Calculus II | Spring and Fall 2026

Teaching Assistant

Summer 2026	Introduction to Probability
Fall 2025	Vector Calculus; Elementary Abstract Algebra I
Spring 2025	Algebra II (graduate); Vector Calculus; Engineering Calculus III
Fall 2024	Algebra I (graduate); Engineering Calculus III
Summer 2024	Precalculus Algebra/Trigonometry
Spring 2024	Elementary Abstract Algebra I; Vector Calculus
Fall 2023	Engineering Calculus III; Elementary Abstract Algebra I
Summer 2023	Precalculus Algebra/Trigonometry
Spring 2023	Engineering Calculus II; Introduction to Discrete Structures
Fall 2022	Calculus I; Calculus III

Honors and Awards

M.V. Johns Jr. Scholarship for Graduate Study in Statistics, 2025 - 2026.
College of Arts and Sciences, University of South Florida.

Fred L. and Helen Tharp Endowed Scholarship, 2024 - 2026.
College of Arts and Sciences, University of South Florida.

Professional Service

Journal referee: Finite Fields and Their Applications; Journal of Number Theory.

Skills

Technical: Wolfram Mathematica, MATLAB, LaTeX; familiarity with C/C++.
Languages: Mandarin (native); English (proficient).